

E2E Automation Framework Use Case

Scenario:

You are working as an SDET in an e-commerce company. Your task is to design and implement an automation framework for:

User logs in → searches product → adds to cart → places order → verifies order confirmation.

Problem Statement:

Design a scalable automation framework using Java + Selenium + TestNG/JUnit.

Functional Requirements:

- Valid & invalid login
- Product search with multiple datasets
- Add to cart and checkout flow
- Order confirmation validation

Framework Requirements:

1. TestNG / JUnit:

- Use annotations (@BeforeMethod, @AfterClass, @Test)
- Assertions
- DataProviders / Parameterization
- Parallel execution
- Test Suite (testng.xml)

2. Maven Project Structure:

- src/main/java → Page classes
- src/test/java → Test classes
- src/test/resources → Test data

3. Logging:

- Log4j / SLF4J
- Log test steps and errors

4. Reporting:

- Extent Reports / Allure
- Include screenshots on failure

5. Design Patterns:

- Page Object Model (POM)
- Page Factory (@FindBy)
- Singleton (WebDriver)
- Factory (Browser setup)
- Fluent design pattern (method chaining)

6. Data-Driven Testing:

- Excel / JSON / CSV inputs

7. Cross-Browser Testing:

- Chrome, Firefox, Edge

8. CI/CD:

- Jenkins or GitHub Actions integration

Expected Deliverables:

- Complete framework
- testng.xml
- Reports and logs
- CI pipeline setup

Bonus:

- Retry mechanism
- Screenshots on failure
- Environment config (QA/UAT/Prod)